CCGL9065: Climate-safe Housing, for the poorest

(Creative common)

Cheung Hoi Shun Marco, 3036396089

It's hot at midnight in the room. The roof of the rooftop unit in Sham Shui Po is like a stove. The air is circulated by a fan. The child isn't reading a climate report. She's envisioning the future of Hong Kong.

Globally, extreme heat is increasing in frequency and intensity, particularly in urban areas; the IPCC reports there is a rise in heatwaves, and heat can be magnified in cities, particularly during nighttime. Hong Kong is particularly susceptible due to its high-rise dwellings, concrete roads and narrow roads, making it very susceptible to the urban heat island effect, along with aircon exhaust. But not everyone feels the heat. Hong Kong has 110,000 subdivided units; 220,000 people live in small rooms that are often poorly ventilated and with little money to cool their homes. Hong Kong will not just test its climate adaptation with seawalls by 2100. It will be with the poor's capacity to live in their apartments. A Climate-Safe Housing Standard is needed in Hong Kong because the poor should not suffer extreme heat because they are poor, live in small areas or the owner didn't put in a window.

The standard should be taken up by the Basic Housing Unit (BHU) system. BHUs currently have minimal standards for subdivided units to be habitable and legally rented, such as size, height, fire safety, water, light and ventilation. That's a start. But "basic" should not be "minimal". It should not be legal to build units with a roof and a toilet. It should be possible to sleep and breathe in the heat.

The Climate-Safe Housing Standard should ensure that all BHUs pass two tests: a heat test and an airflow test. The heat test would demonstrate that it is substantially cooler than the outside peak heat warning temperature. We could make the BHU at least 2°C cooler than the outside peak temperature during the day and cooler than outside temperature on hot nights as a trial standard in Hong Kong. This is not much, but it is significant. It does not promise luxury. It promises no less than the street.

The air flow test would require that all rooms containing a living room or bedroom have at least one external window. An internal, ground floor or corridor facing window should not be sufficient. Have mechanical ventilation, certified to provide at least five air changes per hour, in the unit if this is not possible. The test should be on the property owner's form and be inspected. WHO guidelines on housing associate indoor temperature with health impact, and recommends that ventilation, shading, insulation, orientation and cooling can help to lessen heat stress.

This will help single elderly, kids in subdivided flats, those who work outside and need a cool place at home, and the poor who can't afford to run aircons. Everyone would benefit too, with less peak electricity demand, improved sleep and the end of the ambulance runs. Now it's time for action. Hong Kong should mandate an application for a BHU be accompanied by a Climate-Safe Housing Checklist, which will measure indoor temperatures, must include external windows, be shown to have fresh air, shading, insulation and wiring for safe appliances. Approval should only be granted if the unit is suitable, with landlords to make necessary changes. After approval, home residents should be able to report hot and stuffy BHUs. If the government is to use an approval for a BHU status as a stamp of quality for subdivided flats, heat-safety is a key.

In 2100 Hong Kong may have cut seawalls. But climate adaptation means a more delicate question: Will kids get to sleep on hot summer nights? Climate-proofing begins with climate-proofing homes.